

Problem Set 2

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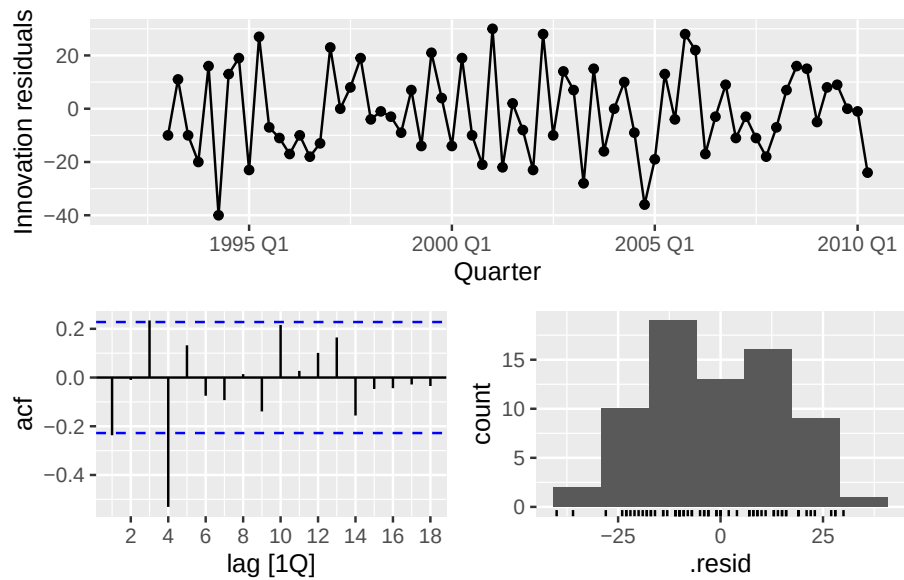
Setup

```
library(fpp3)
```

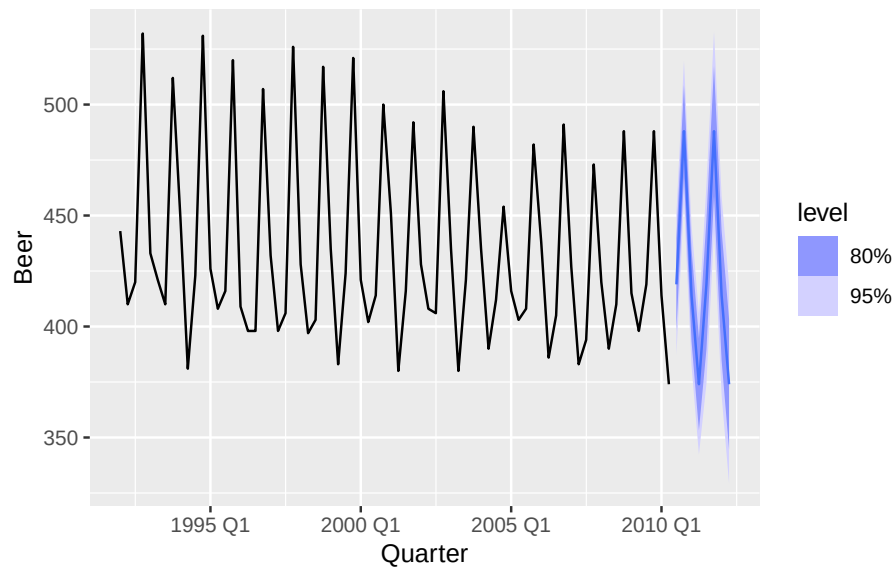
Question 1

Beer from aus_production

```
recent_production <- aus_production %>%  
  filter(year(Quarter) >= 1992)  
  
fit <- recent_production %>%  
  model(SNAIVE(Beer))  
  
fit %>%  
  gg_tsresiduals()
```



```
fit %>%
  forecast() %>%
  autoplot(recent_production)
```



The plot of the forecast looks fairly accurate. Additionally, assesment of the residual plots show that there is no heteroskedasticity. However, there is a significant autocorrelation of the residuals at a lag of 4 quarters, and the residuals are not quite normal since more residuals close to zero was expected. Overall

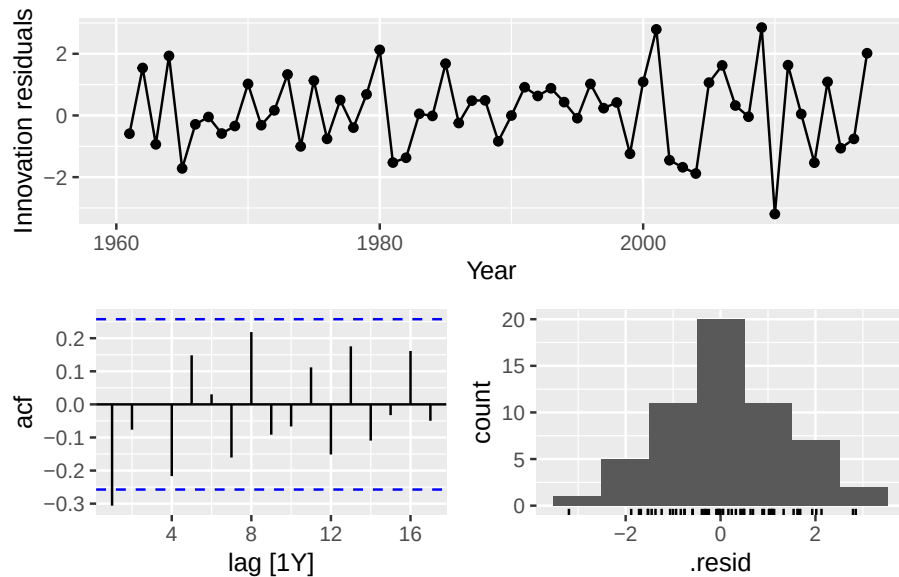
the model isn't bad, but there is likely a better model due to the autocorrelation every 4 quarters. Additionally, the residuals not being perfectly normal is not ideal.

Australian Exports from global_economy

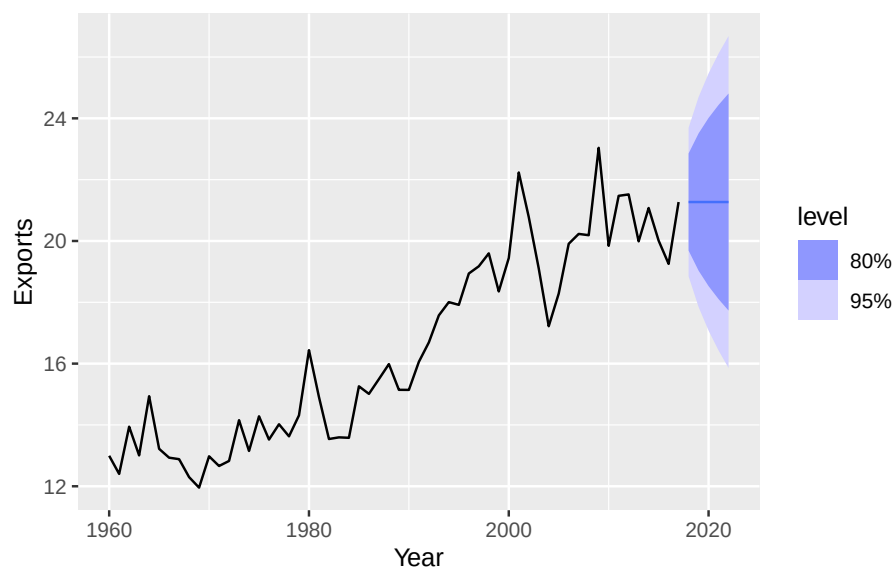
```
australian_economy <- global_economy %>%
  filter(Country == "Australia")

fit <- australian_economy %>%
  model(NAIVE(Exports))

fit %>%
  gg_tsresiduals()
```



```
fit %>%
  forecast(h = "5 years") %>%
  autoplot(australian_economy)
```

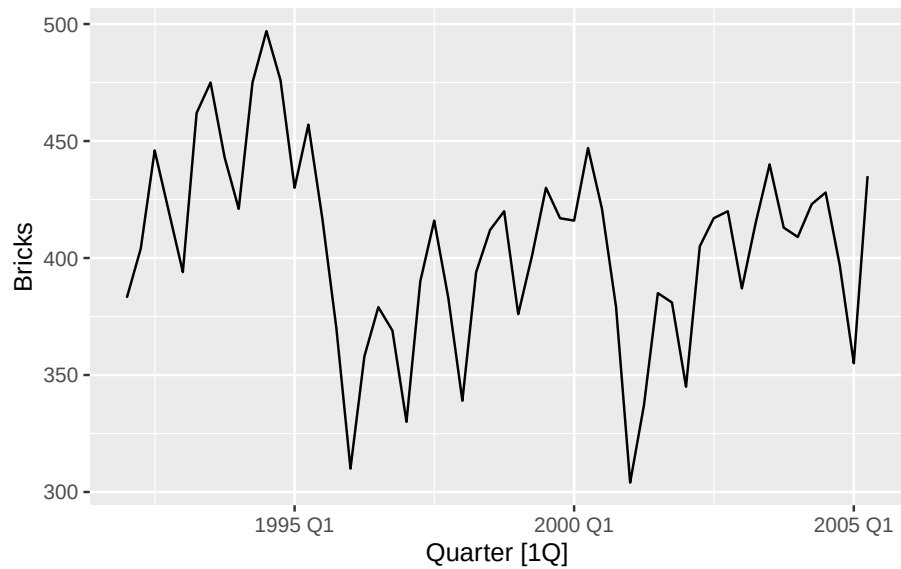


The prediction interval appears quite large which may mean the model is not very useful. However the residual assessment looks fairly good, in that the residuals are homoskedastic, are fairly normal, and only have somewhat significant autocorrelation at lag 1. We can likely trust the model, though it may not show us what we need to know due to the large interval.

Bricks from aus_production

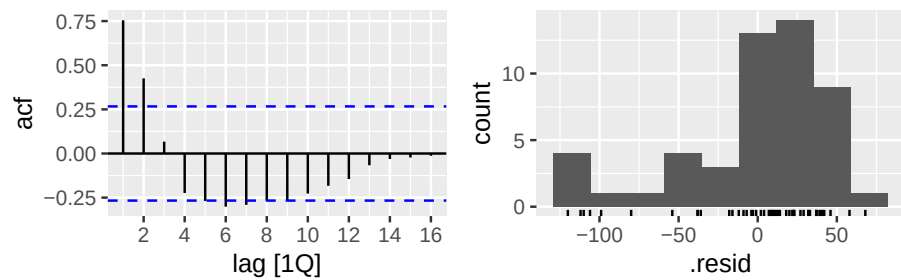
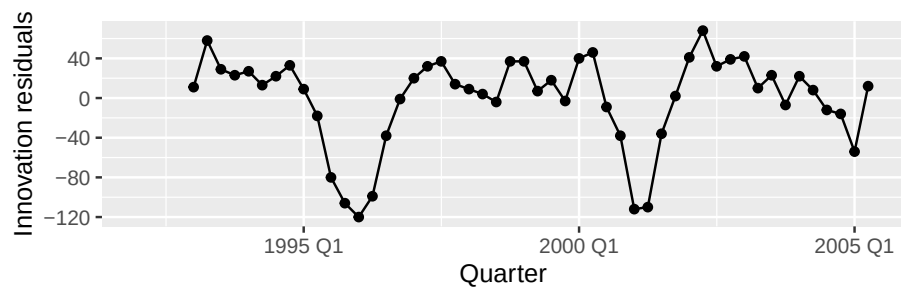
```
recent_production <- aus_production %>%
  filter(year(Quarter) >= 1992) %>%
  select(Quarter, Bricks) %>%
  drop_na()

recent_production %>%
  autoplot(Bricks)
```



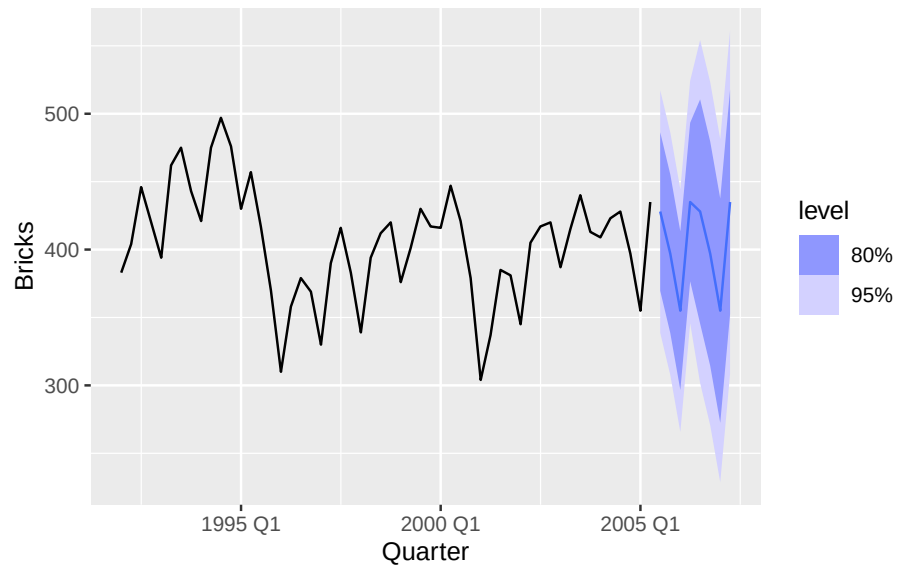
```
fit <- recent_production %>%
  model(SNAIVE(Bricks))

fit %>%
  gg_tsresiduals()
```



```
fit %>%
  forecast(h = "2 years") %>%
```

```
autoplot(recent_production)
```



The forecast captures the seasonal trend well. However, the residual assessment is very poor. The residuals are very autocorrelated and they are also not normal. More refining should be done to reach a model we can trust.

Question 2

Part A: Good forecast methods should have normally distributed residuals.

TRUE. Normality of the residuals indicates the model is behaving as expected and there is a good fit between the model and the data.

Part B: A model with small residuals will give good forecasts.

FALSE. For a single predictor variable the extreme case of this would be a polynomial model with order $n-1$ where n is the number of points. In this case the residuals would be zero but the model would be drastically overfit and likely give very poor forecasts.

Part C: The best measure of forecast accuracy is MAPE.

TRUE. MAPE is the percentage error which makes it very easy to compare across models and applications. “The Best” might be a stretch depending on the situation, but it is very good.

Part D: If your model doesn’t forecast well, you should make it more complicated.

FALSE. Sometimes a more explainable, but worse model can be better. It depends on the situation.

Part E: Always choose the model with the best forecast accuracy as measured on the test set.

FALSE. Again, sometimes a more explainable, but worse model can be better. It depends on the situation.

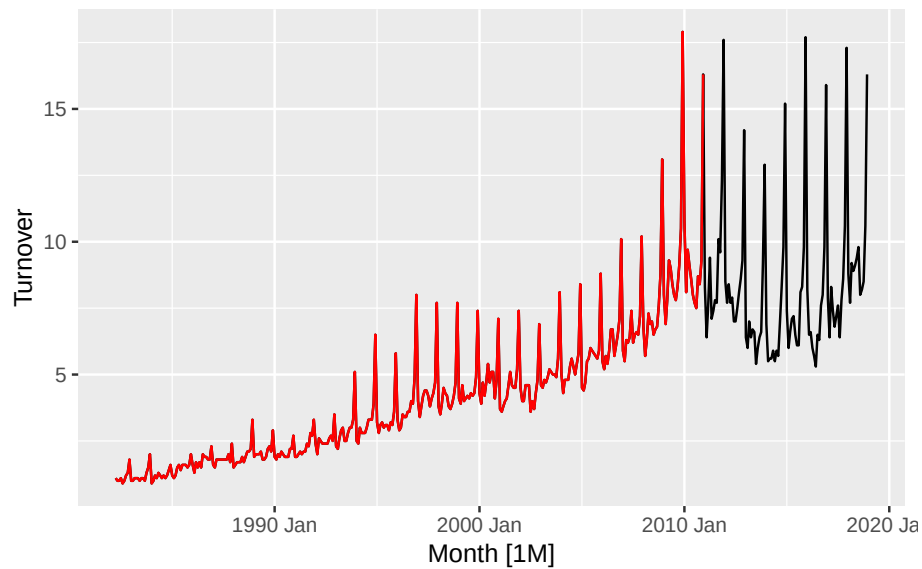
Question 3

```
set.seed(28)
random_series_id = sample(aus_retail$`Series ID`, 1)

series <- aus_retail %>%
  filter(`Series ID` == random_series_id)

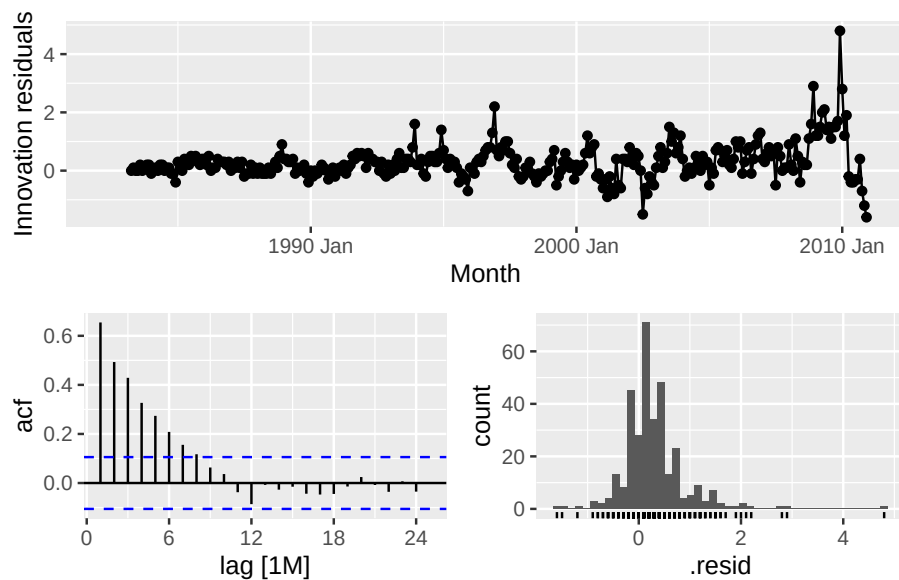
train <- series %>%
  filter(year(Month) < 2011)

series %>%
  autoplot(Turnover) + autolayer(train, Turnover, color = "red")
```



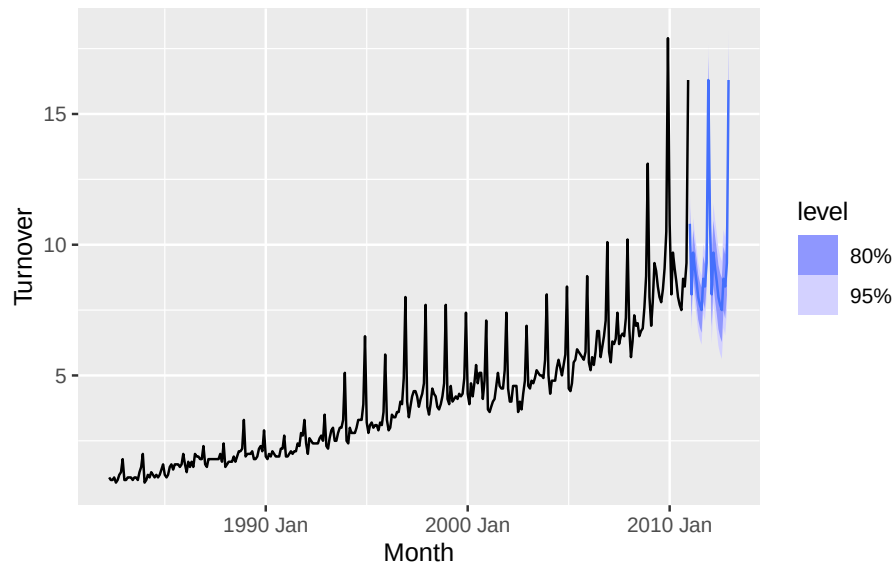
```
fit <- train %>%
  model(SNAIVE(Turnover))

fit %>%
  gg_tsresiduals()
```

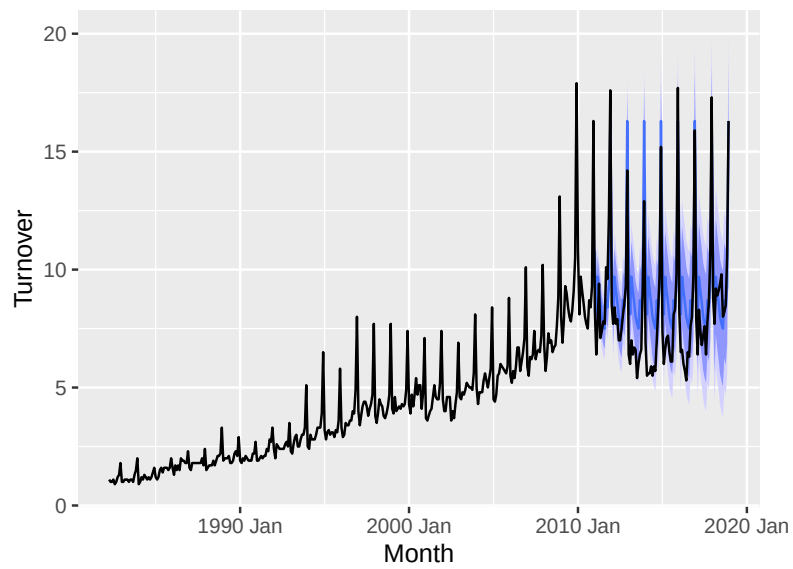
These residual assessment plots show a poor fit between the model and the data. As time goes on the residuals get more extreme showing significant heteroskedasticity. There is also significant autocorrelation between the residuals for all lags up to lag 8. Finally, the residuals appear to have fewer points in the tails than a normal distribution. Overall, this model is likely not very trustworthy based on the residual plots.

```
fit %>%
  forecast() %>%
  autoplot(train)
```



```
forecast <- fit %>%
  forecast(new_data = anti_join(series, train))

forecast %>%
  autoplot(series)
```



```
fit %>%
  accuracy()
```

```
# A tibble: 1 x 12
  State      Industry .model .type    ME  RMSE   MAE   MPE  MAPE  MASE  RMSSE  ACF1
<chr>    <chr>    <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Austral~ Other r~ SNAIV~ Trai~ 0.296 0.675 0.445  6.68  10.4    1      1 0.654

forecast %>%
  accuracy(series)
```

```
# A tibble: 1 x 12
  .model      State Industry .type    ME  RMSE   MAE   MPE  MAPE  MASE  RMSSE  ACF1
<chr>    <chr> <chr>    <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 SNAIVE(T~ Aust~ Other r~ Test -1.10  1.83  1.50 -17.3  20.9  3.36  2.71 0.624
```

The Mean Absolute Percentage Error (MAPE) shows that we are roughly 10.4% off when comparing our model to the training data.

When looking at the accuracy of the model forecast against the test data the MAPE is larger at 20.9%.

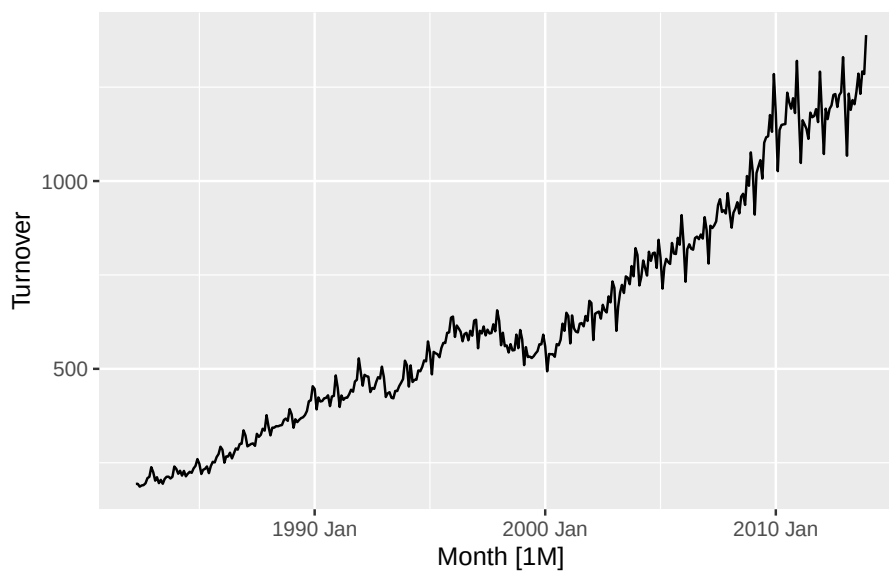
Question 4

Part A

```
takeaway <- aus_retail %>%  
  filter(Industry == "Takeaway food services") %>%  
  summarise(Turnover = sum(Turnover))  
  
max(takeaway$Month)
```

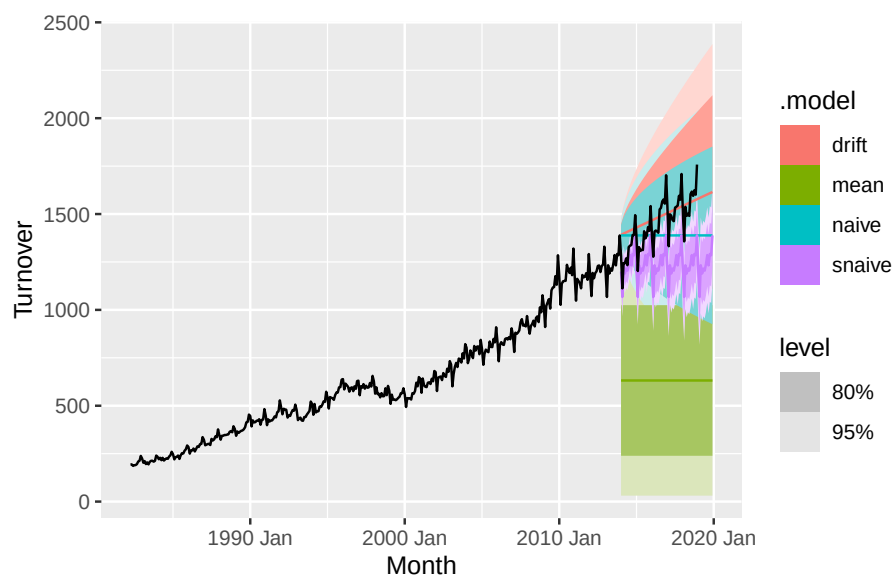
```
<yearmonth[1]>  
[1] "2018 Dec"
```

```
train <- takeaway %>%  
  filter(year(Month) < 2014)  
  
train %>%  
  autoplot(Turnover)
```



Part B

```
fit <- train %>%  
  model(mean = MEAN(Turnover), naive = NAIVE(Turnover), snaive = SNAIVE(Turnover), drift = F)  
  
fit %>%  
  forecast(h = "6 years") %>%  
  autoplot(takeaway)
```



Part C

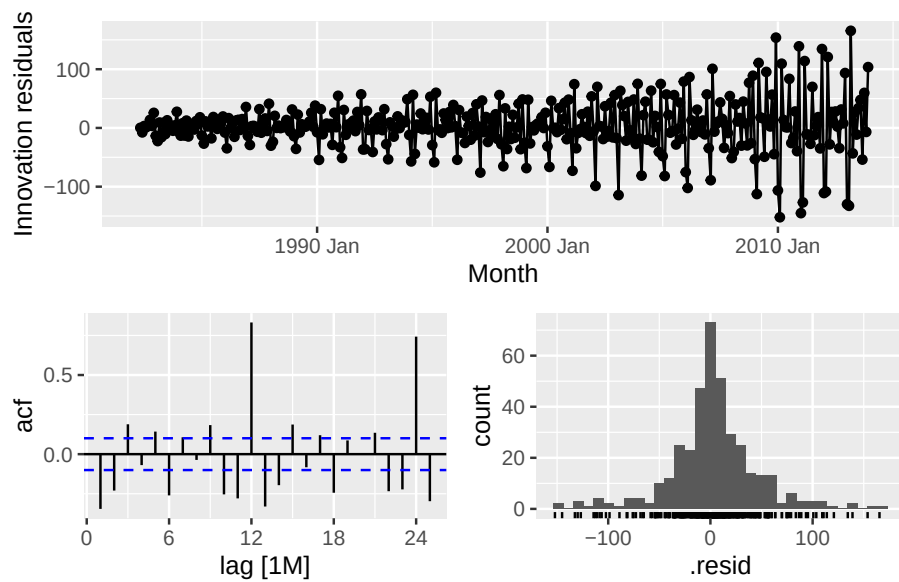
```
fit %>%
  forecast() %>%
  accuracy(takeaway)
```

```
# A tibble: 4 x 10
  .model .type    ME  RMSE  MAE  MPE  MAPE  MASE  RMSSE  ACF1
  <chr>  <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 drift  Test  -92.1  122.  104.  -7.34  8.14  2.54  2.34  0.374
2 mean   Test   704.  710.  704.  52.5  52.5  17.2  13.7  0.494
3 naive  Test  -52.8  106.  82.1  -4.46  6.42  2.00  2.05  0.494
4 snaive Test   99.8  109.  99.8   7.36  7.36  2.43  2.09  0.513
```

The Naive method performed the best with the smallest MAPE at 6.42%. The Mean model is significantly worse than the others.

Part D

```
fit %>%
  select(naive) %>%
  gg_tsresiduals()
```

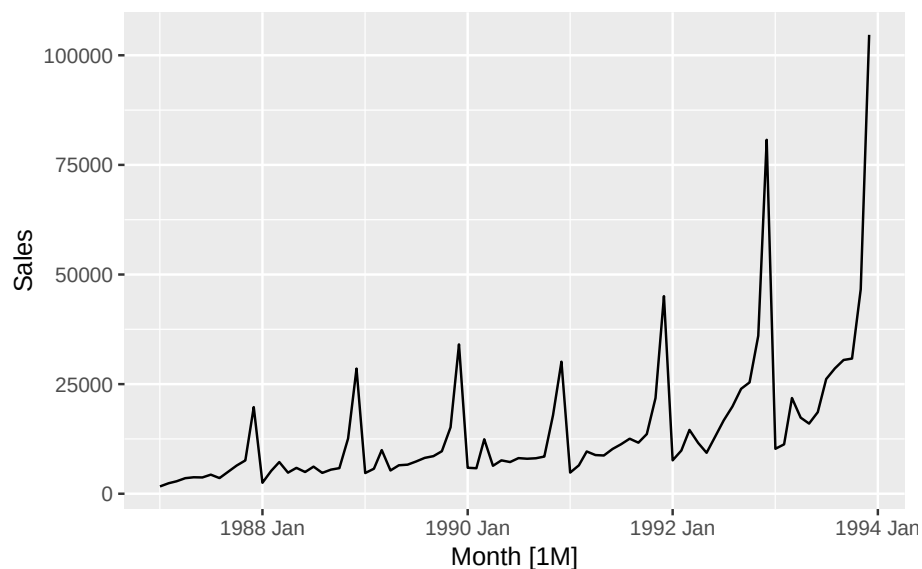


No the residuals for the Naive model do not represent white noise as there is significant heteroskedasticity with the residuals increasing in magnitude as time goes on. This suggests our model does not have an ideal fit.

Question 5

Part A

```
souvenirs %>%  
  autoplot()
```



Part B

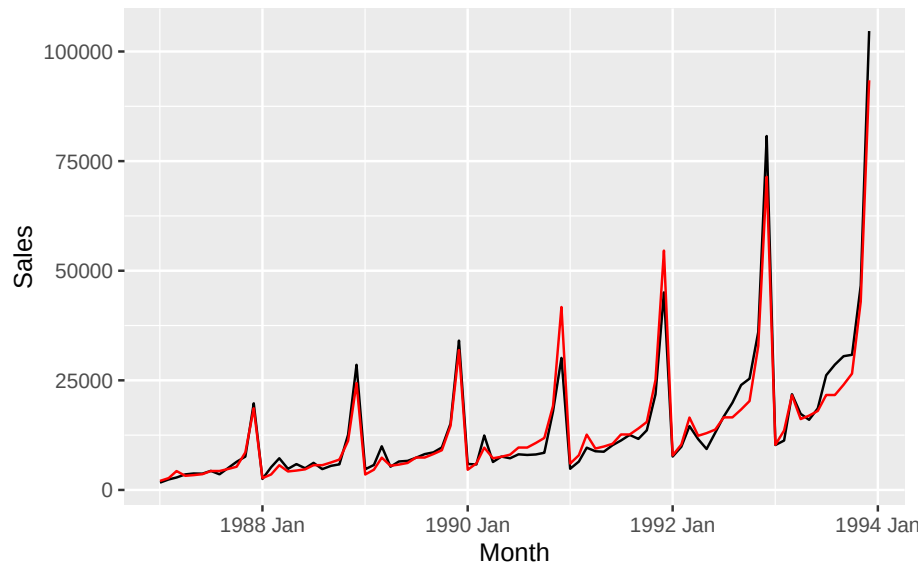
Since there is a non-linear increasing trend taking the log will allow a linear model to be more effective. Log allows for assessment of a multiplicative relationship.

Part C

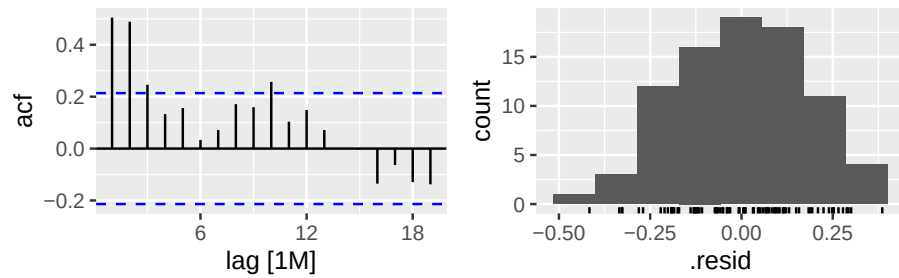
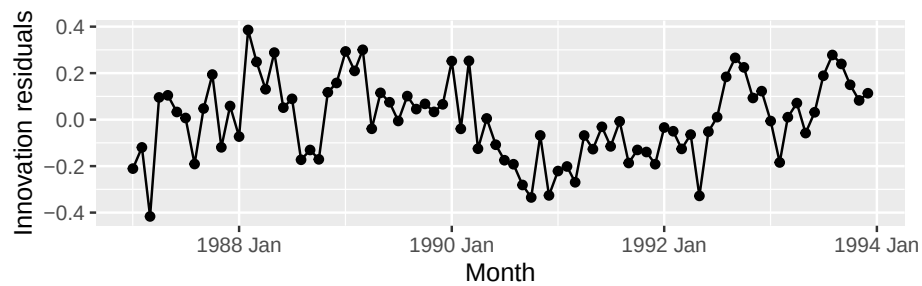
```
df <- souvenirs %>%  
  mutate(festival = if_else(month(Month) == 12, 1, 0))  
  
fit <- df %>%  
  model(TSLM(log(Sales) ~ trend() + season() + festival))
```

Part D

```
fit %>%  
  augment() %>%  
  ggplot(aes(x = Month)) + geom_line(aes(y = Sales), color = "black") + geom_line(aes(y = .f
```



```
fit %>%
  gg_tsresiduals()
```

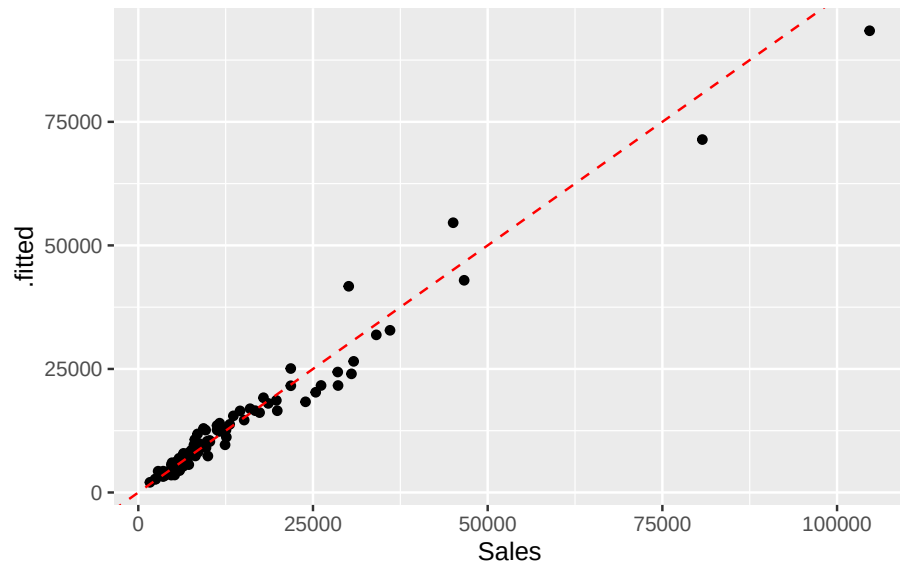


Most concerning here is the statistically significant autocorrelation between the residuals at a lag of 1 and 2.

```
fit %>%
  augment() %>%
```



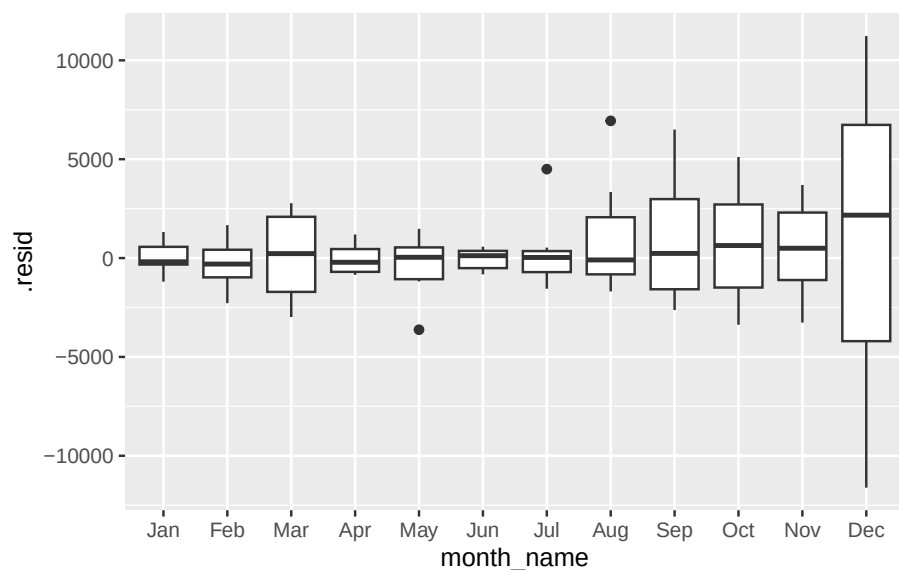
```
ggplot(aes(x = Sales, y = .fitted)) + geom_point() + geom_abline(intercept = 0, slope = 1,
```



This looks decent to me, can still see the autocorrelation though

Part E

```
fit %>%
  augment() %>%
  mutate(month_name = month(Month, label = TRUE)) %>%
  ggplot(aes(x = month_name, y = .resid)) + geom_boxplot()
```



The residuals are not consistent across the months which shows that our model does not properly account for all the seasonality in the data.

Part F

```
fit %>%
  tidy()
```

A tibble: 14 x 6

	.model	term	estimate	std.error	statistic	p.value
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	TSLM(log(Sales) ~ trend() + sea~	(Int~	7.61	0.0769	98.9	8.19e-
78						
2	TSLM(log(Sales) ~ trend() + sea~	tren~	0.0224	0.000845	26.5	1.48e-
38						
3	TSLM(log(Sales) ~ trend() + sea~	seas~	0.251	0.0993	2.53	1.37e- 2
4	TSLM(log(Sales) ~ trend() + sea~	seas~	0.695	0.0993	7.00	1.18e- 9
5	TSLM(log(Sales) ~ trend() + sea~	seas~	0.383	0.0994	3.85	2.52e- 4
6	TSLM(log(Sales) ~ trend() + sea~	seas~	0.408	0.0994	4.11	1.06e- 4
7	TSLM(log(Sales) ~ trend() + sea~	seas~	0.447	0.0994	4.50	2.63e- 5
8	TSLM(log(Sales) ~ trend() + sea~	seas~	0.608	0.0995	6.12	4.69e- 8
9	TSLM(log(Sales) ~ trend() + sea~	seas~	0.585	0.0995	5.88	1.21e- 7
10	TSLM(log(Sales) ~ trend() + sea~	seas~	0.666	0.0996	6.69	4.27e- 9
11	TSLM(log(Sales) ~ trend() + sea~	seas~	0.744	0.0996	7.47	1.61e-
10						
12	TSLM(log(Sales) ~ trend() + sea~	seas~	1.20	0.0997	12.1	7.22e-
19						

```

13 TSLM(log(Sales) ~ trend() + sea~ seas~ 1.96 0.0998 19.6 2.12e-
30
14 TSLM(log(Sales) ~ trend() + sea~ fest~ NA NA NA NA

```

Honestly a bit confused here.

Part G

```

fit %>%
  augment() %>%
  pull(.resid) %>%
  Box.test(lag = 12, type = "Ljung-Box")

```

Box-Ljung test

```

data: .
X-squared = 66.759, df = 12, p-value = 1.287e-09

```

With a very low p-value we can be very confident that the residuals are autocorrelated.

Part H

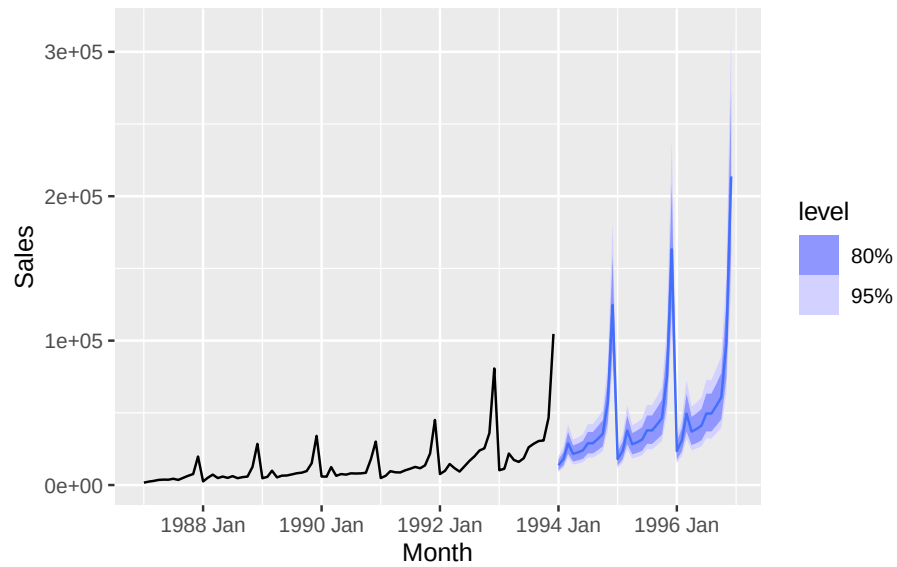
```

future <- df %>%
  new_data(n = 36) %>%
  mutate(festival = if_else(month(Month) == 12, 1, 0))

fc <- fit %>%
  forecast(new_data = future, level = 95)

autoplot(fc) + autolayer(df, Sales)

```



Part H.i

The general view of the model looks decent at extrapolating the observed pattern. I would be skeptical of the actual data reaching those extremes though. Since the residual distributions vary by month, there is additional information in the seasonality of the data that the model is not accounting for.

⑥ Model: $y_t = \beta_0 + \beta_1 t$ with $\sum_{t=1}^T t = \frac{1}{2} T(T+1)$ $\sum_{t=1}^T t^2 = \frac{1}{6} T(T+1)(2T+1)$
 $\hat{\beta} = (X'X)^{-1} X'y$ $\text{Var}(\hat{y}) = \sigma^2 [1 + x^* (X'X)^{-1} (x^*)']$

⑦ $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \\ 1 & T \end{bmatrix}$ from $y_t = \beta_0 + \beta_1 t$.

So $X' = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 2 & \dots & T \end{bmatrix}$.

Then $X'X = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 2 & \dots & T \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \\ 2 \\ \vdots \\ T \end{bmatrix}$
 $= \begin{bmatrix} T & \sum_{t=1}^T t \\ \sum_{t=1}^T t & \sum_{t=1}^T t^2 \end{bmatrix} = \begin{bmatrix} T & \frac{1}{2} T(T+1) \\ \frac{1}{2} T(T+1) & \frac{1}{6} T(T+1)(2T+1) \end{bmatrix}$
 $= \frac{1}{6} \begin{bmatrix} 6T & 3T(T+1) \\ 3T(T+1) & T(T+1)(2T+1) \end{bmatrix}$

⑧ $(X'X)^{-1} = \left(\frac{1}{6}\right)^{-1} \frac{1}{6T^2(T+1)(2T+1) - 9T^2(T+1)^2} \begin{bmatrix} T(T+1)(2T+1) & -3T(T+1) \\ -3T(T+1) & 6T \end{bmatrix}$
 $= \left(\frac{1}{T}\right) \frac{1}{(6T^2 + 6T)(2T+1) - 9T(T^2 + 2T+1)} \begin{bmatrix} T(T+1)(2T+1) & -3T(T+1) \\ -3T(T+1) & 6T \end{bmatrix}$
 $= \frac{12T^3 + 12T^2 + 6T^2 + 6T - 9T^3 - 18T^2 - 9T}{6} \begin{bmatrix} (T+1)(2T+1) & -3(T+1) \\ -3(T+1) & 6 \end{bmatrix}$
 $= \frac{6}{3T^3 - 3T} \begin{bmatrix} (T+1)(2T+1) & -3(T+1) \\ -3(T+1) & 6 \end{bmatrix}$
 $= \frac{2}{T(T^2 - 1)} \begin{bmatrix} (T+1)(2T+1) & -3(T+1) \\ -3(T+1) & 6 \end{bmatrix}$

$$\textcircled{c} \quad \hat{\beta} = (X'X)^{-1} X'y \quad X'y = \begin{bmatrix} 1, & 1, \dots, 1 \\ 1, & 2, \dots, T \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_T \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T t y_t \end{bmatrix}$$

$$= \frac{2}{T(T^2-1)} \begin{bmatrix} (T+1)(2T+1), & -3(T+1) \\ -3(T+1), & 6 \end{bmatrix} \begin{bmatrix} \sum_{t=1}^T y_t \\ \sum_{t=1}^T t y_t \end{bmatrix}$$

$$= \frac{2}{T(T^2-1)} \begin{bmatrix} (T+1)(2T+1) \sum_{t=1}^T y_t - 3(T+1) \sum_{t=1}^T t y_t \\ -3(T+1) \sum_{t=1}^T y_t + 6 \sum_{t=1}^T t y_t \end{bmatrix}$$

$$\textcircled{a} \quad \hat{\beta}_0 = \frac{2}{T(T+1)(T-1)} \left((T+1)(2T+1) \sum_{t=1}^T y_t - 3(T+1) \sum_{t=1}^T t y_t \right)$$

$$= \boxed{\frac{2}{T-1} \left(2T+1 \sum_{t=1}^T y_t - 3 \sum_{t=1}^T t y_t \right)}$$

$$\textcircled{b} \quad \hat{\beta}_1 = \frac{2}{T(T^2-1)} \left(-3(T+1) \sum_{t=1}^T y_t + 6 \sum_{t=1}^T t y_t \right)$$

$$= \boxed{\frac{6}{T(T^2-1)} \left(2 \sum_{t=1}^T t y_t - (T+1) \sum_{t=1}^T y_t \right)}$$

$$\textcircled{1} \quad X^* = [1, T+h] \quad (X^*)' = \begin{bmatrix} 1 \\ T+h \end{bmatrix}$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 [1 + X^* (X'X)^{-1} (X^*)']$$

$$= \sigma^2 \left[1 + [1, T+h] \left(\frac{2}{T(T-1)} \begin{bmatrix} (T+1)(2T+1), & -3(T+1) \\ -3(T+1), & 6 \end{bmatrix} \right) \begin{bmatrix} 1 \\ T+h \end{bmatrix} \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \begin{bmatrix} (T+1)(2T+1) + (T+h)(-3(T+1)), \\ -3(T+1) + (T+h)6 \end{bmatrix} \begin{bmatrix} 1 \\ T+h \end{bmatrix} \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} (T+1) \begin{bmatrix} 2T+1-3T-3h, \\ -3T-3+6T+6h \end{bmatrix} \begin{bmatrix} 1 \\ T+h \end{bmatrix} \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \begin{bmatrix} -1-T-3h, & \frac{3T+6h-3}{T+1} \end{bmatrix} \begin{bmatrix} 1 \\ T+h \end{bmatrix} \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(-1-T-3h + \frac{(3T+6h-3)(T+h)}{T+1} \right) \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(-1-T-3h + \frac{3T^2+6Th-3T+3Th+6h^2-3h}{T+1} \right) \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(-1-T-3h + \frac{-3T^2-3Th-3T-3h+6T^2+12Th+6h^2}{T+1} \right) \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(-1-T-3h + \frac{-3(T+1)(T+h)}{T+1} + \frac{6(T+h)^2}{T+1} \right) \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(-1-T-3h -3T-3h + \frac{6(T+h)^2}{T+1} \right) \right]$$

$$= \sigma^2 \left[1 + \frac{2}{T(T-1)} \left(1-4T-6h + \frac{6(T+h)^2}{T+1} \right) \right]$$